

The Krusell–Smith Economy in Discrete Time: A Master Equation Formulation

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Abstract

Bilal (2023, “Solving Heterogeneous Agent Models with the Master Equation”) develops the Master Equation representation of heterogeneous-agent economies and its first-order approximation (the FAME) in *continuous* time, using the Krusell and Smith (1998) economy as the leading example. This note writes down the *discrete-time* analogue. The logic is unchanged: the value function is defined on the space of distributions and a single Bellman equation summarizes the equilibrium. The only structural differences are mechanical translations of the continuous-time objects: the infinitesimal generator L becomes a one-period transition (Markov) operator $\mathcal{T}$, the discount *rate* ρ becomes a discount *factor* β , the Kolmogorov forward equation $\partial_t g = L^*[g]$ becomes the one-step push-forward $g_{t+1} = \mathcal{T}^*[g_t]$, and “sums of generators” ($L_1 + L_2$) become “products of transition operators” ($\mathcal{T}_1 \mathcal{T}_2$). We verify that the discrete-time Master Equation collapses to Bilal’s continuous-time Master Equation as the period length $\Delta t \rightarrow 0$, and we derive the discrete-time FAME, including aggregate shocks.

Contents

1 Setup

Time is discrete, $t = 0, 1, 2, \dots$. There is a unit mass of households. A household is described by the idiosyncratic state $x = (a, y)$, where $a \geq 0$ are asset holdings (subject to a borrowing limit $a \geq \underline{a}$, normalized to $\underline{a} = 0$) and y is idiosyncratic labor productivity. Let $\mathcal{X} = [0, \infty) \times Y$ denote the idiosyncratic state space.

Idiosyncratic productivity. Productivity y_t is an exogenous, stationary Markov process, independent across households, with transition kernel $P(y, dy') = \Pr(y_{t+1} \in dy' \mid y_t = y)$. Define the associated *income transition operator* (a Markov, i.e. expectation, operator)

$$(P\varphi)(y) \equiv \mathbb{E}[\varphi(y_{t+1}) \mid y_t = y] = \int \varphi(y') P(y, dy').$$

In continuous time Bilal uses the *generator* $M(y)$ of the productivity process (e.g. $M(y)[V] = \mu(y)V'(y) + \frac{\sigma(y)^2}{2}V''(y)$ for a diffusion). Its discrete-time counterpart is the one-period operator P , related to a generator by $P = \text{Id} + \Delta t M + o(\Delta t)$.

Preferences and budget. Households have time-separable preferences with period utility u (e.g. CRRA, $u(c) = \frac{c^{1-\gamma}-1}{1-\gamma}$) and discount factor $\beta \in (0, 1)$. They save in a single risk-free asset and face the flow budget constraint

$$c_t + a_{t+1} = (1 + r_t) a_t + w_t y_t, \quad a_{t+1} \geq 0, \quad (1)$$

where r_t is the net interest rate and w_t the wage. (The discount factor maps to the continuous-time rate by $\beta = e^{-\rho \Delta t}$; the gross return $1 + r_t$ maps to $1 + r_t \Delta t$.)

Technology and prices. A representative firm operates $Y_t = Z K_t^\alpha N_t^{1-\alpha}$, renting capital K_t (equal to aggregate assets) and hiring labor N_t (equal to aggregate efficiency units). With no depreciation, profit maximization and market clearing pin prices to the current distribution g_t of $x = (a, y)$:

$$r_t = R(g_t) \equiv \alpha Z \left(\frac{K(g_t)}{N(g_t)} \right)^{\alpha-1}, \quad w_t = W(g_t) \equiv (1 - \alpha) Z \left(\frac{K(g_t)}{N(g_t)} \right)^\alpha, \quad (2)$$

where $K(g) = \int a g(x) dx$ and $N(g) = \int y g(x) dx$. As in Bilal's equation (2), R and W are functionals of the distribution.

2 Sequential equilibrium

Household problem. Taking the price sequences $\{r_t, w_t\}$ as given, the household solves

$$V_t(a, y) = \max_{a' \geq 0} u((1 + r_t)a + w_t y - a') + \beta \mathbb{E}[V_{t+1}(a', y') | y]. \quad (3)$$

Let $\hat{a}_t(x) \equiv \hat{s}_t(x)$ denote the optimal saving policy and $\hat{c}_t(x) = (1 + r_t)a + w_t y - \hat{s}_t(x)$ optimal consumption. The interior optimality (Euler) condition is

$$u'(\hat{c}_t(x)) = \beta (1 + r_{t+1}) \mathbb{E}[u'(\hat{c}_{t+1}(a', y')) | y], \quad (4)$$

with complementary slackness at the borrowing constraint $a' = 0$.

Controlled transition operator. Bundle the deterministic asset transition $a' = \hat{s}_t(x)$ with the stochastic income transition P into the one-period *controlled transition operator* $\mathcal{T}_t$ acting on functions of x ,

$$(\mathcal{T}_t \varphi)(x) \equiv \mathbb{E}[\varphi(x_{t+1}) | x_t = x] = \int \varphi(\hat{s}_t(x), y') P(y, dy'), \quad (5)$$

the discrete-time analogue of the generator $L_t(x, c)$ in Bilal's equation (1). Its adjoint $\mathcal{T}_t^*$ pushes a distribution one period forward.

Law of motion of the distribution. The continuous-time Kolmogorov forward equation $\partial_t g_t = L_t^*[g_t]$ (Bilal eq. 3) becomes, in discrete time, the one-step *push-forward*

$$g_{t+1}(x') = (\mathcal{T}_t^* g_t)(x') = \int \mathbf{1}\{\hat{s}_t(a, y) = a'\} P(y, dy') g_t(a, y) da. \quad (6)$$

Equation (??) is the discrete-time Kolmogorov forward equation: it simply transports mass according to the saving policy and the income chain.

Equilibrium. Given g_0 , a sequential equilibrium is a path $\{V_t, \hat{s}_t, r_t, w_t, g_t\}$ such that (??) holds with prices (??) and the distribution evolves by (??). As in continuous time, the difficulty is the forward–backward structure: (??) is solved backward, (??) forward, and prices are the fixed point that clears markets each period.

3 The discrete-time Master Equation

Following Bilal, make the distribution an explicit state variable. Substitute prices $r_t = R(g_t)$, $w_t = W(g_t)$ into the Bellman equation and replace the time index by an explicit dependence on the distribution, $V_t(x) \equiv V(x, g_t)$. Knowing the law of motion (??) means the household can forecast next period’s distribution: write the one-step distribution map

$$\Phi(g) \equiv \mathcal{T}^*(\cdot, g)[g], \quad (7)$$

i.e. $\Phi(g)$ is the distribution obtained by applying (??) once, using the policy that is optimal given the current distribution g . Then the household problem becomes fully recursive:

$$\boxed{V(x, g) = \max_{a' \geq 0} u((1 + R(g))a + W(g)y - a') + \beta \int V((a', y'), \Phi(g)) P(y, dy').} \quad (8)$$

Equation (??) is the **discrete-time Master Equation**. It is the exact discrete-time counterpart of Bilal’s continuous-time Master Equation

$$\rho V(x, g) = \max_c u(c) + L(x, c, g)[V] + \int \frac{\partial V}{\partial g}(x, x', g) L^*(x', g)[g] dx'. \quad (\text{Bilal, eq. 7})$$

The two continuation terms of the continuous-time equation—the partial-equilibrium generator $L[V]$ (evolution of *own* state) and the term $\int \partial_g V L^*[g]$ (evolution of the *distribution*)—are jointly encoded in the single discrete-time continuation $\beta \int V((a', y'), \Phi(g)) P(y, dy')$: the conditional expectation over (a', y') captures the evolution of the household’s own state, while the argument $\Phi(g)$ captures the evolution of the aggregate distribution. As in continuous time, (??) is the *only* equation that needs to be solved: the distributional fixed point has been absorbed into the value function, so the representation is block-recursive.

3.1 Continuous-time limit (consistency check)

Let the period length be Δt , set $\beta = e^{-\rho \Delta t}$, scale the budget as a flow $a_{t+\Delta t} = a_t + (r_t a_t + w_t y_t - c_t) \Delta t$, and expand the operators to first order in Δt :

$$\mathcal{T} = \text{Id} + \Delta t L + o(\Delta t), \quad \Phi(g) = g + \Delta t L^*(g)[g] + o(\Delta t).$$

The Master Equation (??) (with flow utility $u(c) \Delta t$) becomes

$$V(x, g) = \max_c u(c) \Delta t + (1 - \rho \Delta t) (\text{Id} + \Delta t L_x) \left[V(x, g) + \Delta t \int \frac{\partial V}{\partial g}(x, \xi, g) L^*(\xi, g)[g] d\xi \right].$$

Cancelling $V(x, g)$, dividing by Δt and letting $\Delta t \rightarrow 0$ recovers exactly Bilal’s equation (7). Thus the discrete-time formulation is consistent with, and reduces to, the continuous-time Master Equation—confirming that “the same methodology naturally works in discrete time.”

4 The discrete-time FAME

We now perturb (??) around a deterministic steady state, following Bilal's First-order Approximation to the Master Equation (FAME).

Steady state. A steady state is $(V^{\text{ss}}, g^{\text{ss}})$ with prices $r^{\text{ss}} = R(g^{\text{ss}})$, $w^{\text{ss}} = W(g^{\text{ss}})$, policy $s^{\text{ss}}(x)$, consumption $c^{\text{ss}}(x)$, solving

$$V^{\text{ss}}(x) = \max_{a' \geq 0} u((1 + r^{\text{ss}})a + w^{\text{ss}}y - a') + \beta (\mathcal{T} V^{\text{ss}})(x), \quad g^{\text{ss}} = \mathcal{T}^* g^{\text{ss}},$$

where $\mathcal{T}, \mathcal{T}^*$ are the steady-state transition operator (??) and its adjoint, evaluated at s^{ss} .

Impulse Value ansatz. Let $h \equiv g - g^{\text{ss}}$ be a small distributional impulse. To first order the value function is linear in h :

$$V(x, g^{\text{ss}} + h) = V^{\text{ss}}(x) + \int v(x, \xi) h(\xi) d\xi + o(\|h\|), \quad (9)$$

where $v(x, \xi) = \frac{\partial V}{\partial g}(x, \xi, g^{\text{ss}})$ is the *Impulse Value*: the general-equilibrium effect on the value of a household at $x = (a, y)$ of adding an infinitesimal mass of households at $\xi = (a_\xi, y_\xi)$. This is identical to the continuous-time object; only the equation it satisfies differs.

Derivation. Substitute (??) into (??) and linearize in h , using the envelope theorem (the saving policy is held at its optimum, so its first-order variation drops out). Three first-order channels appear.

(i) *Direct price impact.* Adding mass at ξ changes K by a_ξ and N by y_ξ , hence moves prices by $dr = R_K a_\xi + R_N y_\xi$ and $dw = W_K a_\xi + W_N y_\xi$ (derivatives of (??) at g^{ss}). The induced change in disposable income, valued at the steady-state marginal utility, is $u'(c^{\text{ss}}(x)) D(x, \xi)$ with the *price-impact function*

$$D(x, \xi) = (R_K a + W_K y) a_\xi + (R_N a + W_N y) y_\xi, \quad (10)$$

exactly Bilal's $D(x, x') = (R_0 a_\xi + R_1 y_\xi) a + (W_0 a_\xi + W_1 y_\xi) y$ with $R_0 = R_K, R_1 = R_N, W_0 = W_K, W_1 = W_N$.

(ii) *Continuation through transitions.* The continuation $\beta \int v(x^+, \xi) d[\Phi]$ expands using $\Phi(g^{\text{ss}} + h) = g^{\text{ss}} + \mathcal{T}^* h + \mathcal{G}h$, where $\mathcal{T}^* h$ is the partial-equilibrium push-forward of the impulse (policies fixed) and $\mathcal{G}$ is the general-equilibrium response of the push-forward to policy changes (defined below). The push-forward part, after taking adjoints, transports the *second* argument of v one period forward; the expectation over the household's own next state transports the *first* argument.

(iii) *General-equilibrium savings response.* The price change makes every household ζ revise its saving by $\delta s(\zeta, \xi)$, which reshapes next period's distribution through the kernel $G(\cdot, \xi)$ of $\mathcal{G}$.

Collecting terms, the **discrete-time deterministic FAME** is

$$\boxed{v(x, \xi) = \underbrace{u'(c^{\text{ss}}(x)) D(x, \xi)}_{\text{direct price impact}} + \beta \underbrace{(\mathcal{T}_1 \mathcal{T}_2 v)(x, \xi)}_{\text{continuation: own state } x \text{ and impulse } \xi} + \beta \underbrace{(\mathcal{T}_1[v \star G])(x, \xi)}_{\text{GE: savings responses of other households}}, \quad (11)$$

where $\mathcal{T}_1$ ($\mathcal{T}_2$) applies the transition operator (??) to the first (second) argument of v , and $(v \star G)(x, \xi) \equiv \int v(x, \zeta) G(\zeta, \xi) d\zeta$.

Comparison with continuous time. Bilal’s continuous-time FAME (his eq. 9) reads

$$\rho v(x, x') = u'(c^{\text{ss}}(x))D(x, x') + L(x)[v(\cdot, x')] + L(x')[v(x, \cdot)] + \int v(x, x'')G(x'', x', v) dx''.$$

The dictionary is exact:

- the discount factor β replaces $1/\rho$ (more precisely $1 - \rho \Delta t$);
- the *product* of transition operators $\mathcal{T}_1 \mathcal{T}_2$ replaces the *sum* of generators $L(x) + L(x')$ (because $\mathcal{T}_1 \mathcal{T}_2 = (\text{Id} + \Delta t L_1)(\text{Id} + \Delta t L_2) = \text{Id} + \Delta t(L_1 + L_2) + o(\Delta t)$);
- the GE kernel G is the discrete-time analogue of $G(x'', x', v)$ and likewise depends on v through the consumption (dMPC) response, so (??) is *quadratic* in v , just like the continuous-time FAME.

The general-equilibrium kernel. The per-period saving response of household ζ to a unit impulse at ξ is the change in resources minus the consumption response,

$$\delta s(\zeta, \xi) = D(\zeta, \xi) - \delta c(\zeta, \xi),$$

where δc is obtained by linearizing the Euler equation (??) (a “distributional MPC”). The kernel G is then the linearized push-forward of the displaced mass: in continuous time $G(\zeta, \xi, v) = -\partial_{a_\zeta}[g^{\text{ss}}(\zeta) \delta s(\zeta, \xi)]$ with $\delta c = \frac{1}{u''(c^{\text{ss}})} \partial_a v$; in discrete time the divergence $-\partial_a(\cdot)$ is replaced by the linearization of the push-forward (??) with respect to the saving policy,

$$(\mathcal{G}h)(x') = \frac{\partial}{\partial g} \left[\mathcal{T}^*(\cdot, g)[g] \right]_{g^{\text{ss}}} \cdot h, \quad (\mathcal{G}h)(x') = \int G(x', \xi) h(\xi) d\xi.$$

4.1 Matrix (Sylvester) form

Discretize x on a grid $\{x_i\}$, write $\mathbf{V} = [v(x_i, x_j)]$, let T be the steady-state transition matrix (row-stochastic), $T^\top$ its transpose, $\tilde{D} = [u'(c_i^{\text{ss}})D(x_i, x_j)]$, and $G = G(\mathbf{V})$ the discretized GE kernel. Because $\mathcal{T}_1$ left-multiplies, $\mathcal{T}_2$ right-multiplies by $T^\top$, and $v \star G$ is right-multiplication by G , the FAME (??) becomes the **nonlinear (Riccati-type) Sylvester equation**

$$\boxed{\mathbf{V} = \tilde{D} + \beta T \mathbf{V} (T^\top + G(\mathbf{V}))}. \quad (12)$$

This is the discrete-time analogue of Bilal’s continuous-time Sylvester equation $\rho v = D + Lv + vL^\top + vGv$ and is solved by the same fast iterative schemes.

4.2 Linearized law of motion of the distribution

Perturbing (??) once v (hence G) is known gives the discrete-time analogue of Bilal’s eq. (10),

$$h_{t+1} = \mathcal{T}^* h_t + \mathcal{G} h_t \quad \Longleftrightarrow \quad h_{t+1} = (T^\top + G) h_t. \quad (13)$$

The first term is the partial-equilibrium transport of the impulse; the second is the general-equilibrium feedback through savings. Iterating (??) forward delivers impulse responses without solving any further fixed point (block recursivity). **Stability** is the discrete-time mirror of the continuous-time condition: convergence back to steady state requires the spectral radius of $T^\top + G$ to be strictly below 1 (rather than the eigenvalues of $L^* + \mathcal{G}$ having strictly negative real part).

5 The Huggett variant: an implicitly priced asset

In Krusell–Smith the asset is productive capital and its price is the marginal product, a *technological* functional of the distribution. Huggett (1993) is the polar case: a pure-exchange economy in which the asset is in fixed net supply and its price is determined *behaviorally*, by market clearing. The Master Equation accommodates this with no change in logic—only the price functional is replaced.

5.1 Setup

There is no production: y is an exogenous *endowment* (still Markov with operator P), so there is no wage. Households trade a one-period bond. Let q_t be the *current bond price*: a unit of the bond bought at t costs q_t and pays one unit of the good at $t + 1$, so

$$\frac{1}{q_t} = 1 + r_t. \quad (14)$$

Let a_t denote the (matured) bond holdings carried into t , each paying one unit, and a_{t+1} the bonds purchased at t . The budget constraint is

$$c_t + q_t a_{t+1} = a_t + y_t, \quad a_{t+1} \geq \underline{a}. \quad (15)$$

Writing the budget with q_t rather than r_t is the economically correct timing: the period- t saving choice trades at the *current* price q_t , whereas the realized return $r_t = 1/q_t - 1$ in (??) is the payoff on *already-held* bonds a_t and is predetermined. Thus r_t does not affect time- t behavior; q_t does, and q_t is the price that clears the market. The bond is in fixed net supply B (pure Huggett sets $B = 0$; $B > 0$ represents government debt).

The asset-demand function. To determine the price, treat the current bond price q as a *free variable* rather than imposing $q = Q(g)$. For a candidate q , let

$$\mathcal{A}(x; q, g) \equiv \arg \max_{a' \geq \underline{a}} u(a + y - q a') + \beta \int V((a', y'), \Phi(g)) P(y, dy') \quad (16)$$

denote the household's bond demand when the *current* trade is at price q and the continuation is taken at the equilibrium law of motion $\Phi(g)$. This is the *asset-demand function*: because q enters the current budget, $\mathcal{A}$ genuinely depends on it, with $\partial_q \mathcal{A} < 0$ in the usual case (demand falls as the bond gets more expensive). The *equilibrium* policy is the special case in which the trade occurs at the market-clearing price,

$$\hat{a}(x, g) = \mathcal{A}(x; Q(g), g),$$

which is why the equilibrium policy $\hat{a}(x, g)$ alone carries no free price: it has $q = Q(g)$ already substituted. The bond price $Q(g)$ is the value that clears the market,

$$\int \mathcal{A}(x; q, g) g(x) dx = B \quad \implies \quad q = Q(g) \text{ (defined implicitly)}. \quad (17)$$

Since $\partial_q \mathcal{A} \neq 0$, (??) is a genuine (one-dimensional) equation in q .

5.2 Master Equation

Nothing changes in the recursive logic: $q = Q(g)$ is a (Fréchet-differentiable) functional of the distribution, so Bilal's Assumption 2 is satisfied with an implicitly defined functional. The discrete-time Master Equation (??) becomes

$$V(x, g) = \max_{a' \geq \underline{a}} u(a + y - Q(g) a') + \beta \int V((a', y'), \Phi(g)) P(y, dy'), \quad (18)$$

with the same one-step distribution map $\Phi(g) = \mathcal{T}^*(\cdot, g)[g]$. Steady state is a pair $(V^{\text{ss}}, g^{\text{ss}})$ with $q^{\text{ss}} = Q(g^{\text{ss}})$ clearing (??).

5.3 FAME

The Impulse Value ansatz (??) and the FAME (??) are unchanged in form. Two things are different relative to Krusell–Smith.

(i) A single bond-price channel. Income is an exogenous endowment, so there is no wage. The only price is $q = Q(g)$, and it enters the household's value only through the cost $q a'$ of new bonds. By the envelope theorem, the direct effect of a price change $dq = Q'(g^{\text{ss}})[\delta_\xi]$ on the value, valued at steady-state marginal utility, gives the price-impact function

$$D(x, \xi) = -\hat{a}^{\text{ss}}(x) Q'(g^{\text{ss}})[\delta_\xi], \quad (19)$$

i.e. a household that is buying bonds ($\hat{a}^{\text{ss}}(x) > 0$) is hurt by a higher bond price, in proportion to its bond purchases. (Contrast Krusell–Smith, where the price impact was proportional to asset *holdings* a and worked through both r and w .) Here $Q'(g^{\text{ss}})[\delta_\xi]$ is the marginal effect on the equilibrium bond price of placing an infinitesimal unit of mass at $\xi = (a_\xi, y_\xi)$.

(ii) The price impact is implicit and endogenous to v . Apply the implicit function theorem to the market-clearing condition (??), differentiating in the direction h . The distribution g enters (??) both directly (as the measure $\int \mathcal{A} g$) and through the continuation $\Phi(g)$ inside $\mathcal{A}$, so

$$Q'(g^{\text{ss}})[h] = - \frac{\int \hat{a}^{\text{ss}}(x) h(x) dx + \int \frac{\partial \mathcal{A}}{\partial g} \cdot h g^{\text{ss}}(x) dx}{\int \frac{\partial \mathcal{A}}{\partial q}(x) g^{\text{ss}}(x) dx}, \quad \hat{a}^{\text{ss}}(x) = \mathcal{A}(x; q^{\text{ss}}, g^{\text{ss}}). \quad (20)$$

The denominator $\int \partial_q \mathcal{A} g^{\text{ss}} < 0$ is the *slope of the aggregate bond-demand curve*; the price moves *more* the *less* elastic aggregate demand is. In the numerator, the first term is the mechanical composition effect (adding mass at ξ contributes $\hat{a}^{\text{ss}}(\xi)$ to demand directly), and the second, $\int \partial_g \mathcal{A} \cdot h g^{\text{ss}}$, is the response of demand to the change in the *future* path of prices that h induces through $\Phi(g)$ —the channel that carries the Impulse Value v into the price. In Krusell–Smith the denominator was instead the closed-form curvature of the marginal product, so $D(x, \xi)$ was known once g^{ss} was known, *independently* of v . In Huggett, $\partial_g \mathcal{A}$ depends on the household's valuation of distributional change—i.e. on v itself, through the dMPC. Hence $D = D(v)$, and the FAME becomes the fixed point

$$\mathbf{V} = \tilde{D}(\mathbf{V}) + \beta T \mathbf{V} (T^\top + G(\mathbf{V})), \quad (21)$$

where now *both* $\tilde{D}$ (via the market-clearing elasticity (??)) and G depend on $\mathbf{V}$. The GE kernel G retains the form of Section 4, with the saving response $\delta s(\zeta, \xi) = D(\zeta, \xi) - \delta c(\zeta, \xi)$ now driven by the single implicit price (??). Everything else—block recursivity, the linearized law of motion (??), and the stability criterion (spectral radius of $T^\top + G$ below 1)—carries over unchanged.

Summary of the contrast. Huggett is nested in the same Master Equation as Krusell–Smith: one passes from one to the other simply by replacing the marginal-product price functional with the market-clearing functional (??). The substantive consequence is that the price impact D is no longer a closed-form steady-state object but is itself part of the FAME fixed point, because the market-clearing price responds to the same household behavior that the Impulse Value encodes.

6 Aggregate shocks

Let aggregate productivity be $Z_t = \bar{Z} e^{\varepsilon z_t}$, with z_t a stationary AR(1),

$$z_{t+1} = \phi z_t + \sigma \eta_{t+1}, \quad |\phi| < 1, \quad \eta_{t+1} \sim (0, 1),$$

and define the aggregate expectation operator $(\mathcal{A}\psi)(z) \equiv \mathbb{E}[\psi(z_{t+1}) \mid z_t = z]$, the discrete-time analogue of the generator $A(z)$ in Bilal’s eq. (11). Prices (??) now depend on z through Z . The **Master Equation with aggregate shocks** is

$$V(x, z, g) = \max_{a' \geq 0} u(c) + \beta \mathbb{E} \left[V((a', y'), z', \Phi(g, z)) \mid y, z \right], \quad (22)$$

where $\Phi(g, z)$ is the one-step distribution map evaluated at current aggregate state z and the expectation is over (y', z') .

First-order expansion. With both the distributional deviation and the shock scale of order ε ,

$$V(x, z, g^{\text{ss}} + \varepsilon h) = V^{\text{ss}}(x) + \varepsilon \left\{ \int v(x, \xi) h(\xi) d\xi + \omega(x, z) \right\} + o(\varepsilon),$$

where the deterministic Impulse Value v still solves the deterministic FAME (??) (certainty equivalence / block recursivity), and $\omega(x, z)$ is the *stochastic Impulse Value*. Substituting into (??) yields the **discrete-time stochastic FAME**

$$\boxed{\omega(x, z) = \underbrace{z \Omega(x) u'(c^{\text{ss}}(x))}_{\text{direct price impact of } z} + \beta \underbrace{(\mathcal{A} \mathcal{T}_1 \omega)(x, z)}_{\text{continuation: own state and } z} + \beta \underbrace{\mathcal{T}_1 [v \star S(\cdot, z)](x)}_{\text{GE: savings responses to the aggregate shock}}, \quad (23)$$

where $\Omega(x) = R_Z a + W_Z y$ collects the direct effect of Z on prices (the analogue of Bilal’s $\Omega(x) = R_2 a + W_2 y$), and $S(\cdot, z)$ is the GE savings response to the aggregate shock. Given v , equation (??) is *linear* in ω (the deterministic FAME was quadratic in v), exactly as in continuous time, and reduces to a linear system after discretization.

Law of motion with aggregate shocks. The linearized distribution evolves as the discrete-time analogue of Bilal’s eq. (15),

$$h_{t+1} = (T^\top + G) h_t + S z_t, \quad (24)$$

so that iterating (??) with $\{z_t\}$ produces aggregate impulse responses and characterizes the stochastic steady state.

7 Summary: the continuous-to-discrete dictionary

Object	Continuous time (Bilal)	Discrete time (this note)
Discounting	rate ρ	factor $\beta = e^{-\rho\Delta t}$
Own-state evolution	generator $L(x, c, g)$	transition operator $\mathcal{T}$, eq. (??)
Income process	generator $M(y)$	Markov operator P
Distribution law of motion	$\partial_t g = L^*[g]$	$g_{t+1} = \mathcal{T}^*[g_t] = \Phi(g_t)$
Master Equation	eq. (7)	eq. (??)
Two continuation terms $L_1 + L_2$	sum of generators	product $\mathcal{T}_1\mathcal{T}_2$
FAME	$\rho v = D + Lv + vL^\top + vGv$	$\mathbf{V} = \tilde{D} + \beta T\mathbf{V}(T^\top + G)$
LoM of impulse	$\partial_t h = L^*h + \mathcal{G}h$	$h_{t+1} = (T^\top + G)h_t$
Stability	$\text{Re eig} < 0$	spectral radius < 1
Stochastic FAME	eq. (14), linear in ω	eq. (??), linear in ω

The discrete-time FAME is precisely the analytic, “linearize-then-discretize” foundation for state-space methods in discrete time (Reiter, 2009; Ahn et al., 2018) and is the natural companion of the discrete-time perturbation techniques of Bhandari et al. (2023) and the sequence-space Jacobian approach of Auclert et al. (2021).